Tribhuvan University

Institute of Science and Technology

Patan Multiple Campus

Lab Assignment On

C-Programming

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/*Write a C program to read two integers and check whether they are equal.*/

```
#include <stdio.h>
int main()
{
	int a, b;

	printf("\nEnter first integer: ");
	scanf("%d",&a);

	printf("\nEnter second integer: ");
	scanf("%d",&b);

	if(a == b) {
		printf("\nGiven integers are equal...");
	} else {
		printf("\nGiven integers are not equal...");
	}
	return 0;
}
```

```
Enter first integer: 2

Enter second integer: 2

Given integers are equal...
-------------------------------
```

/*Write a C program to check whether a given number is even or odd.*/

```
#include<stdio.h>
int main()
{
	int n;

	printf("Enter the number to check: ");
	scanf("%d",&n);

	if(n % 2 == 0)
	{
		printf("Even");
	}
	else
	{
		printf("Odd");
	}
	return 0;
}
```

```
Enter the number to check: 33
Odd
---------------------------------
```

/*Write a C program to check whether a given number is positive or negative.*/

```
#include<stdio.h>
int main()
```

```
{
	int n;

	printf("Enter the number: ");
	scanf("%d",&n);

	if(n > 0) {
		printf("Positive");
	}
	else if(n < 0) {
		printf("Negative");
	}
	else {
		printf("It's Zero.");
	}
	return 0;
}
```

```
Enter the number: -2
Negative
------------------------
```

/*Write a C program to read the value of an integer m and display the value of n as 1 when m is greater than 0, as 0 when m is 0, and as -1 when m is less than 0.*/

```
#include <stdio.h>
int main()
{
	int m,n;

	printf("Enter an integer: ");
	scanf("%d",&m);

	if(m > 0)
	{
		n = 1;
		printf("It's %d",n);
	}
	else if(m < 0) {
		n = -1;
		printf("It's %d",n);

	} else {
	n = 0;
	printf("It's %d",n);
	}
	return 0;
}
```

```
Enter an integer: -3
It's -1
----------------------------
```

/*Write a C program to check whether an alphabet letter is a vowel or consonant.*/

```
#include <stdio.h>
#include <ctype.h>

int main()
{
   char ch;
   char vowels[] = {'a','e','i','o','u'};
   int isVowel = 0;

   printf("Enter an alphabet letter: ");
   scanf("%c", &ch);

   ch = tolower(ch);

   for (int i = 0; i < 5; i++) {
      if (ch == vowels[i]) {
         isVowel = 1;
         break;
      }
   }

   if (ch >= 'a' && ch <= 'z') {
      printf(isVowel ? "It is a vowel.\n" : "It is a consonant.\n");
   } else {
      printf("Invalid input.\n");
   }

   return 0;
}
```

```
Enter an alphabet letter: C
It is a consonant.

--------------------------
```

/*Write a C program that reads the age of a candidate and determines whether they are eligible to cast their vote.*/

```
#include <stdio.h>
int main()
{
         int age,i;
         bool hasVoterId;

         printf("Enter your age: ");
         scanf("%d",&age);

         printf("\nDo you have voter id?(1 for true, 0 for false");
         scanf("%d",&i);
```

```
        if(i == 1) {
                hasVoterId = true;
        } else if(i == 0) {
                hasVoterId = false;
        } else{
                printf("Invalid Condition");
                return 0;
        }

        if(age >= 18) {
                if(hasVoterId) {
                printf("Can Vote...");
                } else {
                        printf("Cannot vote..");
                }
        }
        else {
                printf("Cannot vote.");
        }
return 0;
}
```

```
Enter your age: 18

Do you have voter id?(1 for true, 0 for false0
Cannot vote..
--------------------------------
```

/*Write a C program to check whether a character is an alphabet letter, digit, or special character.*/

```
#include <stdio.h>

int main() {
   char ch;

   printf("Enter a character: ");
   scanf("%c", &ch);

   if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
      printf("'%c' is an Alphabet letter.\n", ch);
   }
   else if (ch >= '0' && ch <= '9') {
      printf("'%c' is a Digit.\n", ch);
   }
   else {
      printf("'%c' is a Special character.\n", ch);
   }

   return 0;
}
```

```
Enter a character: #
'#' is a Special character.

--------------------------------
```

/*Write a C program that reads an integer day number (1-7) and displays the corresponding day name.*/

```
#include <stdio.h>

int main() {
   int day;

   printf("Enter day number (1-7): ");
   scanf("%d", &day);

   switch(day) {
      case 1:
         printf("Sunday\n");
         break;
      case 2:
         printf("Monday\n");
         break;
      case 3:
         printf("Tuesday\n");
         break;
      case 4:
         printf("Wednesday\n");
         break;
      case 5:
         printf("Thursday\n");
         break;
      case 6:
         printf("Friday\n");
         break;
      case 7:
         printf("Saturday\n");
         break;
      default:
         printf("Invalid day number! Please enter 1 to 7.\n");
   }
   return 0;
}
```

```
Enter day number (1-7): 7
Saturday
```

/*Write a C program that reads a single digit and displays it in words.*/

```
#include <stdio.h>

int main() {
   int digit;
```

```
    printf("Enter a single digit (0-9): ");
    scanf("%d", &digit);

    switch(digit) {
        case 0:
            printf("Zero\n");
            break;
        case 1:
            printf("One\n");
            break;
        case 2:
            printf("Two\n");
            break;
        case 3:
            printf("Three\n");
            break;
        case 4:
            printf("Four\n");
            break;
        case 5:
            printf("Five\n");
            break;
        case 6:
            printf("Six\n");
            break;
        case 7:
            printf("Seven\n");
            break;
        case 8:
            printf("Eight\n");
            break;
        case 9:
            printf("Nine\n");
            break;
        default:
            printf("Invalid input! Please enter a digit between 0 and 9.\n");
    }

    return 0;
}
```

```
Enter a single digit (0-9): 3
Three

--------------------------------
```

/*Write a C program to find whether a given year is a leap year or not.*/

```
#include <stdio.h>
int main() {
        int year;
```

```
        printf("Enter the year.");
        scanf("%d", &year);

        if ( ((year % 4 == 0) && (year % 100 != 0)) || (year % 400 == 0) )
        {
                printf("Leap year.");
        } else {
                printf("Not leap year.");
        }
        return 0;
}
```

```
Enter the year.2016
Leap year.
-----------------------
```

/*Write a C program that accepts a person's height in centimeters and categorizes them based on their height.*/

```
#include <stdio.h>

int main() {
   float height;

   printf("Enter height in centimeters: ");
   scanf("%f", &height);

   if (height < 150) {
      printf("The person is Short.\n");
   }
   else if (height >= 150 && height < 165) {
      printf("The person is of Average height.\n");
   }
   else if (height >= 165 && height < 180) {
      printf("The person is Tall.\n");
   }
   else {
      printf("The person is Very Tall.\n");
   }
   return 0;
}
```

```
Enter height in centimeters: 163
The person is of Average height.

------------------------------
```

/*Write a C program to find the largest of three numbers.*/

```
#include <stdio.h>

int main() {
   int num1, num2, num3, largest;

   printf("Enter three numbers: ");
```

```
    scanf("%d %d %d", &num1, &num2, &num3);

    if (num1 >= num2 && num1 >= num3) {
        largest = num1;
    }
    else if (num2 >= num1 && num2 >= num3) {
        largest = num2;
    }
    else {
        largest = num3;
    }
    printf("The largest number is: %d\n", largest);
    return 0;
}
```

```
Enter three numbers: 2
3
1
The largest number is: 3
```

/*Write a C program that accepts a coordinate point in the XY coordinate system and determines which quadrant it lies in.*/

```
#include <stdio.h>

int main() {
    float x, y;

    printf("Enter X and Y coordinates: ");
    scanf("%f %f", &x, &y);

    if (x > 0 && y > 0) {
        printf("The point (%.2f, %.2f) lies in Quadrant I\n", x, y);
    }
    else if (x < 0 && y > 0) {
        printf("The point (%.2f, %.2f) lies in Quadrant II\n", x, y);
    }
    else if (x < 0 && y < 0) {
        printf("The point (%.2f, %.2f) lies in Quadrant III\n", x, y);
    }
    else if (x > 0 && y < 0) {
        printf("The point (%.2f, %.2f) lies in Quadrant IV\n", x, y);
    }
    else if (x == 0 && y != 0) {
        printf("The point lies on the Y-axis\n");
    }
    else if (y == 0 && x != 0) {
        printf("The point lies on the X-axis\n");
    }
    else {
        printf("The point is at the Origin (0,0)\n");
```

```
Enter X and Y coordinates: 3
3
The point (3.00, 3.00) lies in Quadrant I

--------------------------------
```

```
    }
    return 0;
}
```

/* Write a C program to calculate the profit or loss on a transaction. */

```
#include <stdio.h>

int main() {
    float costPrice, sellingPrice, result;

    printf("Enter Cost Price: ");
    scanf("%f", &costPrice);

    printf("Enter Selling Price: ");
    scanf("%f", &sellingPrice);

    if (sellingPrice > costPrice) {
        result = sellingPrice - costPrice;
        printf("Profit = %.2f\n", result);
    }
    else if (costPrice > sellingPrice) {
        result = costPrice - sellingPrice;
        printf("Loss = %.2f\n", result);
    }
    else {
        printf("No Profit No Loss\n");
    }
    return 0;
}
```

```
Enter Cost Price: 1200
Enter Selling Price: 1000
Loss = 200.00

---------------------------
```

/*Write a C program to check whether a triangle can be formed given any three angles.*/

```
#include <stdio.h>

int main() {
    float angle1, angle2, angle3, sum;

    printf("Enter three angles of a triangle: ");
    scanf("%f %f %f", &angle1, &angle2, &angle3);

    sum = angle1 + angle2 + angle3;

    if (sum == 180 && angle1 > 0 && angle2 > 0 && angle3 > 0) {
        printf("Yes, a triangle can be formed.\n");
    } else {
```

```
Enter three angles of a triangle: 60
60
45
No, a triangle cannot be formed.

-----------------------------
```

```
        printf("No, a triangle cannot be formed.\n");
    }
    return 0;
}
```

/*Write a C program to check whether a triangle is equilateral, isosceles, or scalene.*/

```
#include <stdio.h>

int main() {
    float side1, side2, side3;

    printf("Enter the three sides of the triangle: ");
    scanf("%f %f %f", &side1, &side2, &side3);

    if (side1 + side2 > side3 && side2 + side3 > side1 && side1 + side3 > side2) {

        if (side1 == side2 && side2 == side3) {
            printf("The triangle is Equilateral.\n");
        }
        else if (side1 == side2 || side2 == side3 || side1 == side3) {
            printf("The triangle is Isosceles.\n");
        }
        else {
            printf("The triangle is Scalene.\n");
        }

    } else {
        printf("Not a valid triangle.\n");
    }
    return 0;
}
```

```
Enter the three sides of the triangle: 60
60
60
The triangle is Equilateral.

--------------------------------
```

/*Write a C program to calculate the roots of a quadratic equation.*/

```
#include <stdio.h>
#include <math.h>

int main() {
    float a, b, c, discriminant, root1, root2, realPart, imagPart;

    printf("Enter coefficients a, b, c: ");
    scanf("%f %f %f", &a, &b, &c);

    if(a == 0) {
        printf("Not a quadratic equation.\n");
```

```
        return 0;
    }

    discriminant = b*b - 4*a*c;

    if(discriminant > 0) {
        root1 = (-b + sqrt(discriminant)) / (2*a);
        root2 = (-b - sqrt(discriminant)) / (2*a);
        printf("Roots are real and distinct: %.2f and %.2f\n", root1, root2);
    }
    else if(discriminant == 0) {
        root1 = root2 = -b / (2*a);
        printf("Roots are real and equal: %.2f and %.2f\n", root1, root2);
    }
    else {
        realPart = -b / (2*a);
        imagPart = sqrt(-discriminant) / (2*a);
        printf("Roots are complex: %.2f + %.2fi and %.2f - %.2fi\n", realPart, imagPart, realPart,
imagPart);
    }
    return 0;
}
```

```
Enter coefficients a, b, c: 2
3
-3
Roots are real and distinct: 0.69 and -2.19

--------------------------------
```

/*Write a C program to read the roll number, name, and marks of the three subjects of a student, and calculate their total percentage and division.*/

```
#include <stdio.h>

int main() {
    int roll;
    char name[50];
    float marks1, marks2, marks3, total, percentage;

    printf("Enter Roll Number: ");
    scanf("%d", &roll);

    printf("Enter Name: ");
    scanf(" %[^\n]", name);

    printf("Enter marks of three subjects: ");
    scanf("%f %f %f", &marks1, &marks2, &marks3);

    total = marks1 + marks2 + marks3;
```

```
    percentage = (total / 300) * 100;

    printf("\nStudent Details:\n");
    printf("Roll Number: %d\n", roll);
    printf("Name: %s\n", name);
    printf("Total Marks: %.2f\n", total);
    printf("Percentage: %.2f%%\n", percentage);

    if (percentage >= 60) {
        printf("Division: First Division\n");
    }
    else if (percentage >= 50) {
        printf("Division: Second Division\n");
    }
    else if (percentage >= 40) {
        printf("Division: Third Division\n");
    }
    else {
        printf("Division: Fail\n");
    }
    return 0;
}
```

```
Enter Roll Number: 12
Enter Name: Mukunda Rijyal
Enter marks of three subjects: 12 12 12

Student Details:
Roll Number: 12
Name: Mukunda Rijyal
Total Marks: 36.00
Percentage: 12.00%
Division: Fail

--------------------------------
```

```
/*Write a C program to find the eligibility for admission to a professional course based on the
following criteria:
Marks in maths >= 65
Marks in physics >= 55
Marks in chemistry >=50
Total marks in above three subjects >= 180
OR
Total marks in maths and other subjects >= 140*/

#include <stdio.h>

int main() {
    float maths, physics, chemistry, otherSubjects, total1, total2;

    printf("Enter marks in Mathematics: ");
    scanf("%f", &maths);

    printf("Enter marks in Physics: ");
    scanf("%f", &physics);

    printf("Enter marks in Chemistry: ");
    scanf("%f", &chemistry);
```

```
    printf("Enter marks in other subjects: ");
    scanf("%f", &otherSubjects);

    total1 = maths + physics + chemistry;
    total2 = maths + otherSubjects;

    if ((maths >= 65 && physics >= 55 && chemistry >= 50 && total1 >= 180)
        || (total2 >= 140)) {
        printf("The candidate is eligible for admission.\n");
    } else {
        printf("The candidate is NOT eligible for admission.\n");
    }
    return 0;
}
```

```
Enter marks in Mathematics: 80
Enter marks in Physics: 70
Enter marks in Chemistry: 75
Enter marks in other subjects: 90
The candidate is eligible for admission.

--------------------------------
```

/*Write a C program to read the temperature in centigrade and display a suitable message according to the given criteria:
Temperature < 0 : Freezing weather
Temperature ranges from 0-10 : Very cold weather
Temperature ranges from 10-20 : Cold weather
Temperature ranges from 20-30 : Normal weather
Temperature ranges from 30-40 : Hot weather
Temperature >= 40 : Very hot weather */

```
#include <stdio.h>

int main() {
    float temp;

    printf("Enter temperature in centigrade: ");
    scanf("%f", &temp);

    if (temp < 0) {
        printf("Freezing weather");
    }
    else if (temp >= 0 && temp < 10) {
        printf("Very cold weather");
    }
    else if (temp >= 10 && temp < 20) {
        printf("Cold weather");
    }
    else if (temp >= 20 && temp < 30) {
        printf("Normal weather");
    }
    else if (temp >= 30 && temp < 40) {
        printf("Hot weather");
```

```
    }
    else {
        printf("Very hot weather");
    }
    return 0;
}
```

```
Enter temperature in centigrade: 37
Hot weather
--------------------------------
```

/*Write a C program to calculate and print the electricity bill for a given customer. Take the customer ID, name, and units consumed as input from the keyboard, then display the total amount due. Use the following charge rates:

Unit	Charge per unit
Up to 199	@1.20
200+ but <400	@1.50
400+ but <600	@1.80
600+	@2.00

If the bill exceeds Rs. 400, a surcharge of 15% will be charged,
and the minimum bill should be Rs. 100.*/

```
#include <stdio.h>

int main() {
    int customer_id;
    char customer_name[50];
    int units;
    float charge, bill, surcharge = 0;

    printf("Enter Customer ID: ");
    scanf("%d", &customer_id);

    printf("Enter Customer Name: ");
    scanf("%s", customer_name);

    printf("Enter Units Consumed: ");
    scanf("%d", &units);

    if (units <= 199) {
        charge = units * 1.20;
    }
    else if (units >= 200 && units < 400) {
        charge = units * 1.50;
    }
    else if (units >= 400 && units < 600) {
        charge = units * 1.80;
    }
    else {
        charge = units * 2.00;
```

```
    }

    if (charge > 400) {
        surcharge = charge * 0.15;
    }

    bill = charge + surcharge;

    if (bill < 100) {
        bill = 100;
    }

    printf("\nElectricity Bill");
    printf("\nCustomer ID   : %d", customer_id);
    printf("\nCustomer Name : %s", customer_name);
    printf("\nUnits Consumed: %d", units);
    printf("\nTotal Amount  : Rs. %.2f", bill);

    return 0;
}
```

```
Enter Customer ID: 122
Enter Customer Name: MukuDon
Enter Units Consumed: 12

Electricity Bill
Customer ID   : 122
Customer Name : MukuDon
Units Consumed: 12
Total Amount  : Rs. 100.00
--------------------------------
```

/*Write a C program to display the first 10 natural numbers.*/

```
#include <stdio.h>
int main()
{
        int i;
        for(i = 0; i < 10; i++)
        {
                printf("%d ",i+1);
        }
        return 0;
}
```

```
1 2 3 4 5 6 7 8 9 10
----------------------
```

/*Write a C program to display n terms of natural numbers and their sum.*/

```
#include <stdio.h>
int main()
{
        int n, i, sum = 0;

        printf("Enter n terms: ");
        scanf("%d",&n);

        for(i = 1; i <= n; i++)
        {
```

```
        printf("%d, ",i);
        sum += i;
    }
    printf("\nThe sum of %d natural numbers is: %d",n,sum);
    return 0;
}
```

```
Enter n terms: 3
1, 2, 3,
The sum of 3 natural numbers is: 6
--------------------------------
```

/* Write a C program to display n terms of even natural numbers and their sum. */

```
#include <stdio.h>

int main()
{
   int i, n, sum = 0;

   printf("Enter n terms: ");
   scanf("%d", &n);

   for(i = 1; i <= n; i++)
   {
      printf("%d ", 2 * i);
      sum += 2 * i;
   }

   printf("\nSum of first %d even natural numbers is: %d", n, sum);
   return 0;
}
```

```
Enter n terms: 3
2 4 6
Sum of first 3 even natural numbers is: 12
--------------------------------
```

/*Write a C program to display the multiplication table of a given integer.*/

```
#include <stdio.h>
int main()
{
    int n, i;

    printf("Enter a number: ");
    scanf("%d",&n);

    for(i = 1; i <= 10; i++)
    {
        printf("%d * %d = %d\n",n,i,n*i);
    }
    return 0;
}
```

```
Enter a number: 2
2 * 1 = 2
2 * 2 = 4
2 * 3 = 6
2 * 4 = 8
2 * 5 = 10
2 * 6 = 12
2 * 7 = 14
2 * 8 = 16
2 * 9 = 18
2 * 10 = 20

--------------------
```

/* Write a C program to display n terms of square natural numbers and their sum */

```
#include <stdio.h>

int main()
{
   int i, n;
   int sum = 0;

   printf("Enter n terms: ");
   scanf("%d", &n);

   for(i = 1; i <= n; i++)
   {
      printf("%d ", i * i);
      sum += i * i;
   }

   printf("\nSum of first %d square natural numbers is: %d", n, sum);
   return 0;
}
```

```
Enter n terms: 3
1 4 9
Sum of first 3 square natural numbers is: 14
--------------------------------
```

/* Write a C program to display the cube of the numbers up to a given integer */

```
#include <stdio.h>

int main()
{
   int i, n;

   printf("Enter a number: ");
   scanf("%d", &n);

   for(i = 1; i <= n; i++)
   {
      printf("Number = %d, Cube = %d\n", i, i * i * i);
   }
   return 0;
}
```

```
Enter a number: 3
Number = 1, Cube = 1
Number = 2, Cube = 8
Number = 3, Cube = 27

--------------------------
```

/* Write a C program to read 10 numbers from the keyboard and find their sum and average */

```
#include <stdio.h>

int main()
{
```

```
    int i;
    float num, sum = 0, average;

    printf("Enter 10 numbers:\n");

    for(i = 1; i <= 10; i++)
    {
        scanf("%f", &num);
        sum += num;
    }

    average = sum / 10;

    printf("Sum = %.2f\n", sum);
    printf("Average = %.2f\n", average);

    return 0;
}
```

```
Enter 10 numbers:
2
1
2
3
1
2
3
4
5
6
Sum = 29.00
Average = 2.90
--------------------
```

/* Write a C program to display n terms of the harmonic series and their sum */

```
#include <stdio.h>

int main()
{
    int i, n;
    double sum = 0.0;

    printf("Enter n terms: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {
        printf("1/%d", i);

        if(i != n)
            printf(" + ");

        sum += 1.0 / i;
    }
    printf("\nSum of harmonic series up to %d terms = %.4lf", n, sum);
    return 0;
}
```

```
Enter n terms: 3
1/1 + 1/2 + 1/3
Sum of harmonic series up to 3 terms = 1.8333
------------------------------
```

/* Write a C program to display the first n terms of Fibonacci series */

```
#include <stdio.h>

int main()
{
   int n, i;
   int first = 0, second = 1, next;

   printf("Enter number of terms: ");
   scanf("%d", &n);

   printf("Fibonacci Series:\n");

   for(i = 1; i <= n; i++)
   {
      printf("%d ", first);

      next = first + second;
      first = second;
      second = next;
   }
   return 0;
}
```

```
Enter number of terms: 6
Fibonacci Series:
0 1 1 2 3 5
--------------------------
```

/* Write a C program to calculate the factorial of a given number */

```
#include <stdio.h>

int main()
{
   int n, i;
   unsigned long long factorial = 1;

   printf("Enter a number: ");
   scanf("%d", &n);

   if(n < 0)
   {
      printf("Factorial of negative numbers is not defined.");
   }
   else
   {
      for(i = 1; i <= n; i++)
      {
         factorial *= i;
```

```
    }
    printf("Factorial of %d = %llu", n, factorial);
  }
  return 0;
}
```

```
Enter a number: 4
Factorial of 4 = 24
----------------------
```

/* Write a C program to display a given number in reverse */

```
#include <stdio.h>

int main()
{
  int n, remainder;
  int reverse = 0;

  printf("Enter a number: ");
  scanf("%d", &n);

  while(n != 0)
  {
    remainder = n % 10;
    reverse = reverse * 10 + remainder;
    n = n / 10;
  }
  printf("Reversed number = %d", reverse);
  return 0;
}
```

```
Enter a number: 678
Reversed number = 876
------------------------
```

/* Write a C program to check whether a number is a palindrome */

```
#include <stdio.h>

int main()
{
  int n, original, remainder;
  int reverse = 0;

  printf("Enter a number: ");
  scanf("%d", &n);

  original = n;

  while(n != 0)
  {
    remainder = n % 10;
    reverse = reverse * 10 + remainder;
```

```
        n = n / 10;
    }
    if(original == reverse) {
                printf("%d is a Palindrome number.", original);
    } else {
        printf("%d is NOT a Palindrome number.", original);
    }
    return 0;
}
```

```
Enter a number: 44
44 is a Palindrome number.
```

/* Write a C program to convert a decimal number into binary without using an array */

```
#include <stdio.h>

int main()
{
    int n, remainder;
    long long binary = 0;
    long long place = 1;

    printf("Enter a decimal number: ");
    scanf("%d", &n);

    while(n > 0)
    {
        remainder = n % 2;
        binary += remainder * place;
        place *= 10;
        n = n / 2;
    }
    printf("Binary number = %lld", binary);
    return 0;
}
```

```
Enter a decimal number: 212
Binary number = 11010100
```

/* Write a C program to determine whether a given number is prime */

```
#include <stdio.h>

int main()
{
    int n, i, isPrime = 1;

    printf("Enter a number: ");
    scanf("%d", &n);

    if(n <= 1)
```

```
    {
        isPrime = 0;
    }
    else
    {
        for(i = 2; i * i <= n; i++)
        {
            if(n % i == 0)
            {
                isPrime = 0;
                break;
            }
        }
    }
    if(isPrime) {
        printf("%d is a prime number.", n);
    } else {
        printf("%d is NOT a prime number.", n);
    }
            return 0;
}
```

```
Enter a number: 3
3 is a prime number.
--------------------
```

/* Write a C program to check whether a number is perfect or not */

```
#include <stdio.h>

int main()
{
    int n, i, sum = 0;

    printf("Enter a number: ");
    scanf("%d", &n);

    for(i = 1; i <= n / 2; i++)
    {
        if(n % i == 0)
            sum += i;
    }
    if(sum == n) {
        printf("%d is a Perfect number.", n);
    } else {
        printf("%d is NOT a Perfect number.", n);
    }
    return 0;
}
```

```
Enter a number: 6
6 is a Perfect number.
----------------------
```

```
/* Write a C program to check whether a number is an Armstrong number */

#include <stdio.h>
#include <math.h>

int main()
{
   int n, original, remainder, sum = 0;
   int digits = 0;

   printf("Enter a number: ");
   scanf("%d", &n);

   original = n;

   int temp = n;
   while(temp != 0)
   {
      digits++;
      temp /= 10;
   }
   temp = n;

   while(temp != 0)
   {
      remainder = temp % 10;
      sum += pow(remainder, digits);
      temp /= 10;
   }

   if(sum == original) {
      printf("%d is an Armstrong number.", original);
   } else {
      printf("%d is NOT an Armstrong number.", original);
   }
   return 0;
}
```

```
Enter a number: 3
3 is an Armstrong number.
--------------------------
```

```
/* Write a C program to find the prime numbers within a given range */

#include <stdio.h>

int main()
{
   int low, high, i, j, isPrime;
```

```
    printf("Enter lower bound: ");
    scanf("%d", &low);
    printf("Enter upper bound: ");
    scanf("%d", &high);

    printf("Prime numbers between %d and %d are:\n", low, high);

    for(i = low; i <= high; i++)
    {
        if(i <= 1) continue;

        isPrime = 1;

        for(j = 2; j * j <= i; j++)
        {
            if(i % j == 0)
            {
                isPrime = 0;
                break;
            }
        }
        if(isPrime) {
            printf("%d ", i);
        }
    }
    return 0;
}
```

```
Enter lower bound: 2
Enter upper bound: 8
Prime numbers between 2 and 8 are:
2 3 5 7
--------------------------------
```

```
/* Write a C program to find perfect numbers within a given range */

#include <stdio.h>

int main()
{
    int low, high, i, j, sum;

    printf("Enter lower bound: ");
    scanf("%d", &low);
    printf("Enter upper bound: ");
    scanf("%d", &high);

    printf("Perfect numbers between %d and %d are:\n", low, high);

    for(i = low; i <= high; i++)
    {
        sum = 0;
```

```
        for(j = 1; j <= i / 2; j++)
        {
            if(i % j == 0)
                sum += j;
        }

        if(sum == i && i != 0) {
            printf("%d ", i);
        }
    }
    return 0;
}
```

```
Enter lower bound: 0
Enter upper bound: 100
Perfect numbers between 0 and 100 are:
6 28
--------------------------------
```

```
/* Write a C program to find Armstrong numbers within a given range */

#include <stdio.h>
#include <math.h>

int main()
{
    int low, high, num, original, remainder, digits;
    int sum;

    printf("Enter lower bound: ");
    scanf("%d", &low);
    printf("Enter upper bound: ");
    scanf("%d", &high);

    printf("Armstrong numbers between %d and %d are:\n", low, high);

    for(num = low; num <= high; num++)
    {
        original = num;
        sum = 0;
        digits = 0;

        int temp = num;
        while(temp != 0)
        {
            digits++;
            temp /= 10;
        }

        temp = num;
```

```
        while(temp != 0)
        {
            remainder = temp % 10;
            sum += pow(remainder, digits);
            temp /= 10;
        }

        if(sum == original) printf("%d ", num);
    }
    return 0;
}
```

```
Enter lower bound: 10
Enter upper bound: 500
Armstrong numbers between 10 and 500 are:
153 370 371 407
--------------------------------
```

```
/* Write a C program to find the sum of the series 1 + 11 + 111 + ... n terms */

#include <stdio.h>

int main()
{
    int n, i;
    long long term = 1;
    long long sum = 0;

    printf("Enter number of terms: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {
        sum += term;
        term = term * 10 + 1;
    }
    printf("Sum of the series is: %lld", sum);
    return 0;
}
```

```
Enter number of terms: 4
Sum of the series is: 1234
----------------------------
```

```
/*Write a C program to find the sum of the series 1-x^2/2!+x^4/4!- .....*/

#include <stdio.h>
#include <math.h>

int main()
{
    int n, i, sign;
    double x, term, sum = 0;

    printf("Enter the value of x: ");
    scanf("%lf", &x);
```

```
    printf("Enter the numbr of terms: ");
    scanf("%d", &n);

    sum = 0;

    for(i = 0; i < n; i++)
    {
        int power = 2 * i;
        long long fact = 1;

        for(int j = 1; j <= power; j++)
        {
            fact *= j;
        }

        if(i % 2 == 0) {
            sign = 1;
        } else {
            sign = -1;
            }
        term = sign * pow(x, power) / fact;
        sum += term;
    }
    printf("Sum of the series = %.6lf", sum);
    return 0;
}
```

```
Enter the value of x: 4
Enter the number of terms: 4
Sum of the series = -2.022222
--------------------------------
```

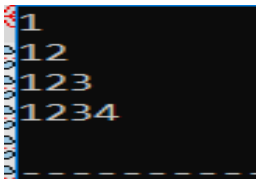
```
/*Write a C program to display the following right-angled triangle pattern using asterisks:
*
**
***
****        */

#include <stdio.h>
int main()
{
    int i, j, n = 4;
    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= i; j++)  {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

```
*
**
***
****
```

/*Write a C program to display the following right-angled triangle pattern with numbers:

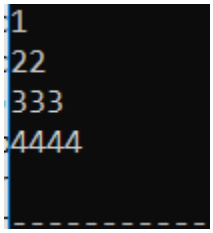
1
12
123
1234 */

```
#include <stdio.h>
int main()
{
   int i, j, n = 4;

   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= i; j++)
      {
         printf("%d", j);
      }
      printf("\n");
   }
   return 0;
}
```



/*Write a C program to display the following right-angled triangle patterns with repeating numbers:
1
22
333
4444 */

```
#include <stdio.h>
int main()
{
   int i, j, n = 4;

   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= i; j++)
      {
         printf("%d", i);
      }
      printf("\n");
   }
   return 0;
}
```

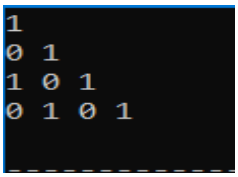


```
/*Write a C program to print Floyd's triangle:
1
2 3
4 5 6
7 8 9 10   */

#include <stdio.h>
int main()
{
   int i, j, n = 4, num = 1;

   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= i; j++)
      {
         printf("%d ", num);
         num++;
      }
      printf("\n");
   }
   return 0;
}
```

```
1
2 3
4 5 6
7 8 9 10
```

```
/*Write a C program to print the following pattern:
1
0 1
0 1 0 1
1 0 1 0 1  */

#include <stdio.h>
int main()
{
   int i, j, n = 4;
   int num;

   for(i = 1; i <= n; i++)
   {
      if(i % 2 == 1)
         num = 1;
      else
         num = 0;

      for(j = 1; j <= i; j++)
      {
         printf("%d ", num);
         num = 1 - num;
```

```
        }
        printf("\n");
    }
    return 0;
}
```



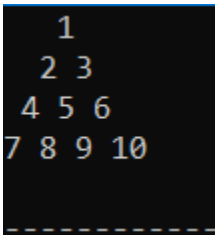
/*Write a C program to display the following pyramid pattern using asterisks:

```
  *
 * *
* * *
* * * *    */

#include <stdio.h>
int main()
{
    int i, j, n = 4;

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
        {
            printf(" ");
        }
        for(j = 1; j <= i; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

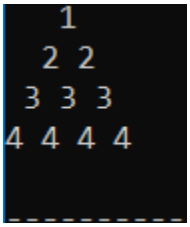
/*Write a C program to display the following pyramid pattern of numbers (increasing by 1):

```
  1
 2 3
4 5 6
7 8 9 10   */

#include <stdio.h>
int main()
{
    int i, j, n = 4, num = 1;

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
```

```
        {
            printf(" ");
        }
        for(j = 1; j <= i; j++)
        {
            printf("%d ", num);
            num++;
        }
        printf("\n");
    }
    return 0;
}
```



```
/*Write a C program to display the following pyramid pattern with repeating numbers:
   1
  2 2
 3 3 3
4 4 4 4  */

#include <stdio.h>
int main()
{
    int i, j, n = 4;

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
        {
            printf(" ");
        }
        for(j = 1; j <= i; j++)
        {
            printf("%d ",i);
        }
        printf("\n");
    }
    return 0;
}
```



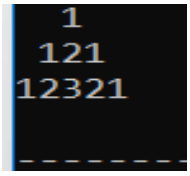
```
/*Write a C program to display Pascal's triangle:
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
*/
```

```c
#include <stdio.h>
int main()
{
   int i, j, n = 5;
   int num;

   for(i = 0; i < n; i++)
   {
      for(j = 0; j < n - i - 1; j++)
      {
         printf(" ");
      }

      num = 1;

      for(j = 0; j <= i; j++)
      {
         printf("%d ", num);
         num = num * (i - j) / (j + 1);
      }
      printf("\n");
   }
   return 0;
}
```

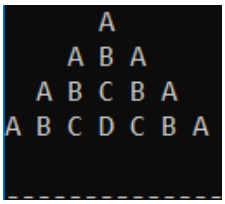
```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

```c
/*Write a C program to display the following pattern for n rows (starting with 1, with 1s at start
and end of each row):
  1
 121
12321   */

#include <stdio.h>
int main()
{
   int i, j, n = 3;

   for(i = 1; i <= n; i++)
   {

      for(j = 1; j <= n - i; j++)
      {
         printf(" ");
      }
      for(j = 1; j <= i; j++)
      {
         printf("%d", j);
```

```
        }

        for(j = i - 1; j >= 1; j--)
        {
            printf("%d", j);
        }
        printf("\n");
    }
    return 0;
}
```



```
/*Write a C program to display the following pattern:
     A
   A B A
 A B C B A
A B C D C B A    */

#include <stdio.h>
int main()
{
    int i, j, n = 4;

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
        {
            printf("  ");
        }
        for(j = 0; j < i; j++)
        {
            printf("%c ", 'A' + j);
        }
        for(j = i - 2; j >= 0; j--)
        {
            printf("%c ", 'A' + j);
        }
        printf("\n");
    }
    return 0;
}
```



```
/*Write a C program to display the following diamond pattern:
   *
  ***
 *****
*******
*********
*******
 *****
  ***
   *                 */

#include <stdio.h>
int main()
{
   int i, j, n = 5;

   for(i = 1; i <= n; i++)
   {
      for(j = 1; j <= n - i; j++)
      {
         printf(" ");
      }

      for(j = 1; j <= 2*i - 1; j++)
      {
         printf("*");
      }
      printf("\n");
   }

   for(i = n - 1; i >= 1; i--)
   {
      for(j = 1; j <= n - i; j++)
      {
         printf(" ");
      }
      for(j = 1; j <= 2*i - 1; j++)
      {
         printf("*");
      }
      printf("\n");
   }
   return 0;
}
```

```
   *
  ***
 *****
*******
*********
*******
 *****
  ***
   *
```